

Datasets Being Used

Peripheral blood Datasets (3):

1) Title: Local and systemic responses to SARS-CoV-2 infection in children and adults

Found: Human Cell Atlas

<https://explore.data.humancellatlas.org/projects/1538d572-bcb7-426b-8d2c-84f3a7f87bb0>

Paper that uses it: <https://www.nature.com/articles/s41586-021-04345-x>

Cell count: Before QC: 137271 after QC: 137124

Number of donors: 26 healthy child donors; Total within dataset n = 93

Ages of donors: Newborn, Infant, 2–5 year-old child stage, 6–12 year-old child stage, Adolescent stage

Populations defined by the paper or data annotations:

Controls:

QC:

Counts per cell mostly above 200: yes

Mitochondrial % acceptable: yes

Healthy-cell profile present (low Mitochondrial, low ribosomal): yes

Read depth near 8,000–10,000: yes

At least 100 cells per cell type: yes

Normalized: Yes ; log1p

2) Title: Human Immune Health Atlas

Found: Chan Zuckerberg CELLxGENE Discover

<https://cellxgene.cziscience.com/collections/77f9d7e9-5675-49c3-abed-ce02f39eef1b>

Paper: <https://www.nature.com/articles/s41586-025-09686-5>

Cell count: 207225 (pediatric healthy data ONLY) After QC: 206865

Number of donors: Total 108 ; 13 donors who are under 18

Ages of donors: 11 (n=10) & 12 (n=3) years old

Populations defined by the paper or data annotations:

Controls:

QC:

Counts per cell mostly above 200: yes

Mitochondrial % acceptable: yes

Healthy-cell profile present (low Mitochondrial, low ribosomal): yes

Read depth near 8,000–10,000: yes

At least 100 cells per cell type: yes

Normalized:

3) **Title:** CITEseq of JDM PBMCs

Found: Chan Zuckerberg CELLxGENE Discover

<https://cellxgene.cziscience.com/collections/c672834e-c3e3-49cb-81a5-4c844be4a975>

Paper: <https://insight.jci.org/articles/view/176963>

Cell count: 23161 (pediatric healthy data ONLY) After QC: 223013

Number of donors: Total 20; 5 healthy donors

Ages of donors: 1–5 year-old child stage (n=3), 6–12 year-old child stage (n=1), 16 year old stage (n=1)

Populations defined by the paper or data annotations:

Controls:

QC:

Counts per cell mostly above 200: yes

Mitochondrial % acceptable: yes

Healthy-cell profile present (low Mitochondrial, low ribosomal): yes

Read depth near 8,000–10,000: yes

At least 100 cells per cell type: yes

Normalized:

Thymus Dataset – SC-RNA:

3) **Title:** Integrated scRNA-Seq Identifies Human Postnatal Thymus Seeding Progenitors and Regulatory Dynamics of Differentiating Immature Thymocyte

Found: Chan Zuckerberg CELLxGENE Discover

<https://cellxgene.cziscience.com/collections/83ed3be8-4cb9-43e6-9aaa-3fbbf5d1bd3a>

Paper: [https://www.cell.com/immunity/fulltext/S1074-7613\(20\)30132-1?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS1074761320301321%3Fshowall%3Dtrue](https://www.cell.com/immunity/fulltext/S1074-7613(20)30132-1?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS1074761320301321%3Fshowall%3Dtrue)

Cell count: 71732

Number of donors: Total 5; all healthy

Ages of donors: 5 month old child stage (n=1), 6 month old stage (n=2), 10 month old stage (n=1), 14 month old stage (n=1)

Controls:

QC:

Counts per cell mostly above 200: yes

Mitochondrial % acceptable: yes

Healthy-cell profile present (low Mitochondrial, low ribosomal): yes

3) **Title:** A multimodal single-cell atlas of the adolescent brain reveals gene regulatory networks linking development to disease risk

Found: Chan Zuckerberg CELLxGENE Discover

<https://cellxgene.cziscience.com/collections/4624894f-6cf7-45d2-91ff-51fff9da9f13>

Paper: <https://www.biorxiv.org/content/10.64898/2026.02.20.707029v1>

Cell count:

Number of donors:

Ages of donors: 5 month old child stage (n=1), 6 month old stage (n=2), 10 month old stage (n=1), 14 month old stage (n=1)

Controls:

QC:

Counts per cell mostly above 200: yes

Mitochondrial % acceptable: yes

Healthy-cell profile present (low Mitochondrial, low ribosomal): yes

Thyroid:

3) **Title:** Single-cell RNA Sequencing Reveals the Heterogeneity in Differentiation Trajectory and Tumor Microenvironment Leading to More Aggressive Phenotypes of Papillary Thyroid Cancer in Children and Young Adult Patients

Found: GEO

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE281736>

Paper: <https://advanced.onlinelibrary.wiley.com/doi/10.1002/adv.202417672>

Cell count:

Number of donors: 5 children/young adult donors

Ages of donors: five children/adolescent/young-adult (CAYA) adjacent normal thyroid samples from tissue region. Exact numeric ages were not provided in the GEO metadata.

Controls:

QC:

Counts per cell mostly above 200: yes

Mitochondrial % acceptable: yes

Healthy-cell profile present (low Mitochondrial, low ribosomal): yes

Growth plate scrna